Computer Science GCE A Level 2026 - Paper 2

Duration: 2h30 | Cameroon GCE Board

General Certificate of Education Board - General Certificate of Education Examination. Computer Science 2 (0795), ADVANCED LEVEL, June 2026. Duration: two and a half hours.

Answer any SIX questions. All questions carry 17 marks each. For your guidance, the approximate mark for each part of a question is indicated in brackets. You are reminded of the necessity for good English and orderly presentation in your answers. In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

Exercise 1 (17 marks)

i. (a) What is cache memory? How is it different from main memory? (3 marks)

(b) Describe **two** other factors that can affect the performance of a CPU. (4 marks)

ii. Consider the truth table below for a digital circuit with three inputs A, B and C, and one output Q.

A	B	C	Q
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

(a) From the truth table, determine the sum of products form for output Q. Hence, show that $Q = A \oplus B \oplus C$. (4 marks)

(b) Ignoring production cost, state **two** advantages of designing the digital circuit using the expression $Q = A \oplus B \oplus C$ rather than the sum of products form obtained in ii(a) above. (2 marks)

(c) Draw a logic circuit diagram for the Boolean expression $F = A \cdot B + (A \oplus B \oplus C)$. (4 marks)

Exercise 2 (17 marks)

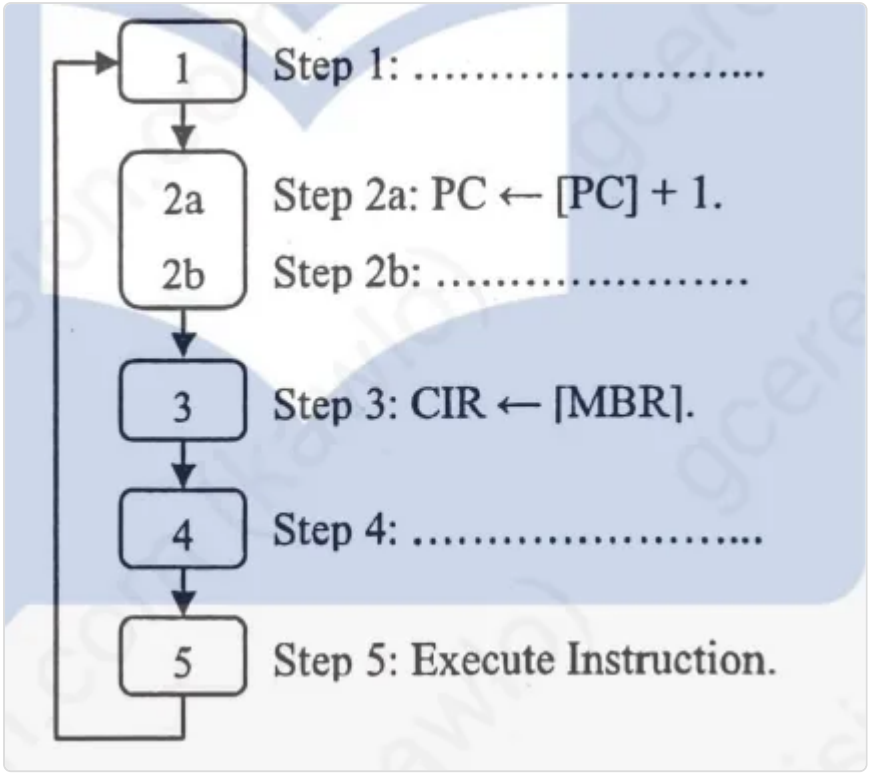
i. The binary pattern 1000 1100 0100 could be interpreted in different ways.

(a) Determine its hexadecimal representation. (2 marks)

(b) Determine its value in denary if it represents an unsigned fixed-point number with four bits after the binary point. (3 marks)

(c) Determine its value in denary if it represents a normalized floating-point number with an 8-bit mantissa and a 4-bit exponent, both in two's complement. (4 marks)

ii. The figure below shows various steps in the fetch-execute cycle. PC, CIR and MBR are processor registers.



(a) State the function of each of the registers PC, CIR and MBR. (3 marks)

(b) Describe the missing steps 1, 2b and 4 using either register transfer notation or a written description. Steps 2a and 2b occur at the same time. (3 marks)

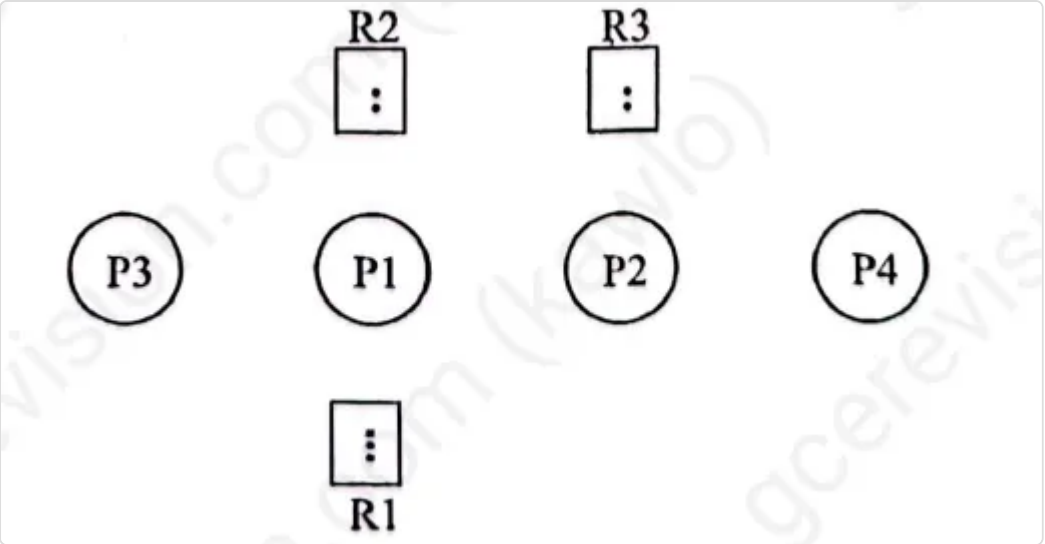
(c) State two examples when the status register might have a bit set. (2 marks)

Exercise 3 (17 marks)

i. A system has four processes P1, P2, P3, and P4. There are three types of dedicated resources R1, R2, and R3 of 3, 2 and 2 units respectively.

- Process P1 holds one unit of R1 and requests one unit of R2.
- Process P2 holds two units of R2 and requests two units of R1 and one unit of R3.
- Process P3 holds one unit of R1 and requests one unit of R2.
- Process P4 holds two units of R3 and requests one unit of R1.

(a) Copy and complete the resource allocation graph below by inserting requests and assignment edges where applicable. (5 marks)



(b) What is a deadlock? State whether the system in (i)(a) above is deadlocked or not. (3 marks)

(c) Describe how deadlocks can be prevented by removing the "*hold and wait" and "non-preemption*" conditions. (2 marks)

(d) Why can't the mutual exclusion condition be eliminated completely when preventing deadlocks? (1 mark)

ii. Context switching is an essential feature of multitasking operating systems.

(a) Explain what the term *multitasking* means. (1 mark)

(b) Why is context switching a necessary feature in multitasking operating systems? (2 marks)

(c) Explain briefly how context switching allows for multitasking. (3 marks)

Exercise 4 (17 marks)

i. Describe the role of the following devices during data transmission.

(a) Repeater.

(b) Multiplexer.

(c) Gateway. (3 marks)

ii. (a) Draw a well labeled diagram to show the different components in any data communication system. (4 marks)

(b) Give **two** main reasons why most organisations prefer client-server networks to peer-to-peer networks. (4 marks)

iii. The principal of a school decides to connect the school's local area network to the Internet and then set up a Wi-Fi connection in order to allow wireless access to the Internet for teachers on campus.

(a) Identify **three** devices that will be required to support these changes. (3 marks)

(b) On a specific day, the Wi-Fi transmission is significantly slower than on previous days. Describe **three** factors that could have caused this degradation in transmission speed. (3 marks)

Exercise 5 (17 marks)

i. A social networking site uses a database to store the details of the people who have joined the site. When a person joins the site, he/she needs to enter some personal data which are only stored after validation.

(a) What is validation as used in this context? (1 mark)

(b) Describe **one** rule each that can be used to validate a person's *name*, *gender*, *password* and *email address*. Use an alternative rule for each data. (4 marks)

ii. Consider the following relational database schema where the fields `StudentId` and `SubjectCode` in the table `Results` are derived from the key fields in the tables `Students` and `Subjects` respectively, and referential integrity is enforced.

```
CREATE TABLE Students(
  StudentId    int NOT NULL,
  StudentName  varchar(30),
  DateOfBirth  date/time,
  PRIMARY KEY(StudentId));

CREATE TABLE Subjects(
  SubjectCode  int NOT NULL
  SubjectTitle varchar(20),
  PRIMARY KEY(SubjectCode));

CREATE TABLE Results(
  StudentId    int NOT NULL,
  SubjectCode  int NOT NULL,
  Grade        char,
  PRIMARY KEY(StudentId,SubjectCode)
  FOREIGN KEY StudentId REFERENCES
    Students(StudentId)
  FOREIGN KEY SubjectCode REFERENCES
    Subjects(SubjectCode));
```

(a) Explain what the term referential integrity means. (2 marks)

(b) Present exactly **one** row of sample data for the `Students` table and **two** rows of sample data for each of the tables `Subjects` and `Results`. (3 marks)

(c) Draw an entity-relationship (E-R) diagram that shows the degrees of the relationships that exist between the three tables above. [*Note: attributes are not required.*] (3 marks)

(d) Write an SQL command which uses the **Inner Join** to retrieve the IDs and names of all the students who have an 'A' grade in the subject *"Computer Science"*. (4 marks)

Exercise 6 (17 marks)

i. Study the pseudocode below and answer the questions that follow.

```
get number
for i ← 2 to number do
  num ← 2
```

```
  check ← *"Yes"*
  while num * num <= i do
    if i mod num = 0 then
      check ← *"No"*
    endif
    num ← num + 1
  endwhile
  if check = *"Yes"* then
    print(i)
  endif
endfor
```

(a) What is the data type of the variable `check`? (1 mark)

(b) Give a trace of the above algorithm for `number` = 10. (4 marks)

(c) State what the algorithm above computes. (1 mark)

(d) Give a flowchart representation of the while loop in the algorithm above. (4 marks)

ii. Iteration and recursion are different ways to execute a set of instructions repeatedly.

(a) Describe how each of iteration and recursion achieves repetition. (2 marks)

(b) State when each of iteration and recursion does not terminate. (2 marks)

(c) Why is iteration faster than recursion? (3 marks)

Exercise 7 (17 marks)

i. (a) Why would an organisation consider building its own software even when it is cheaper to buy one? (2 marks)

(b) What is the purpose of the system development life cycle (SDLC)? (2 marks)

(c) What is the difference between the pilot implementation methodology and the parallel implementation methodology? (2 marks)

(d) State and describe **two** objectives of the design phase of the SDLC. (4 marks)

ii. (a) State **two** ways through which confidentiality of information can be achieved. (2 marks)

(b) Give an example of a scenario where availability can be compromised. (2 marks)

(c) State **three** characteristics of information that make it useful for decision-making. (3 marks)

Exercise 8 (17 marks)

i. (a) Describe how an imperative programming language executes. State **one** example of imperative programming language. (3 marks)

(b) What is variable declaration and what type of programming error arises when one fails to declare a variable. (2 marks)

(c) State **two** advantages and **two** disadvantages of programming using imperative high-level languages as opposed to low-level languages. (4 marks)

ii. Consider the functions Enigma and Mystery below, where m, n, x and y are positive integers and mod is an operator that returns the remainder of a division.

```
function enigma(m, n) ≡
  if (n = 0) then
    return m
  else
    return enigma(n, m mod n)
  endif
endfunction

function mystery(x, y) ≡
  if (x = 0 or y = 0) then
    return 0
  else
    return (x * y) / enigma(x, y)
  endif
endfunction
```

(a) Determine the values of `enigma(4, 6)` and `mystery(6, 4)`. (4 marks)

(b) What does each of the functions Enigma and Mystery compute? (2 marks)

(c) Which of the above functions is recursive? State the reason. (2 marks)

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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